

Ans. to the ques. no. 1

4bit number highest is 1111 which is 15. So the numbers

exist from 0-15. Max number for output =  $15+15=30$ . ( $A+B$ )

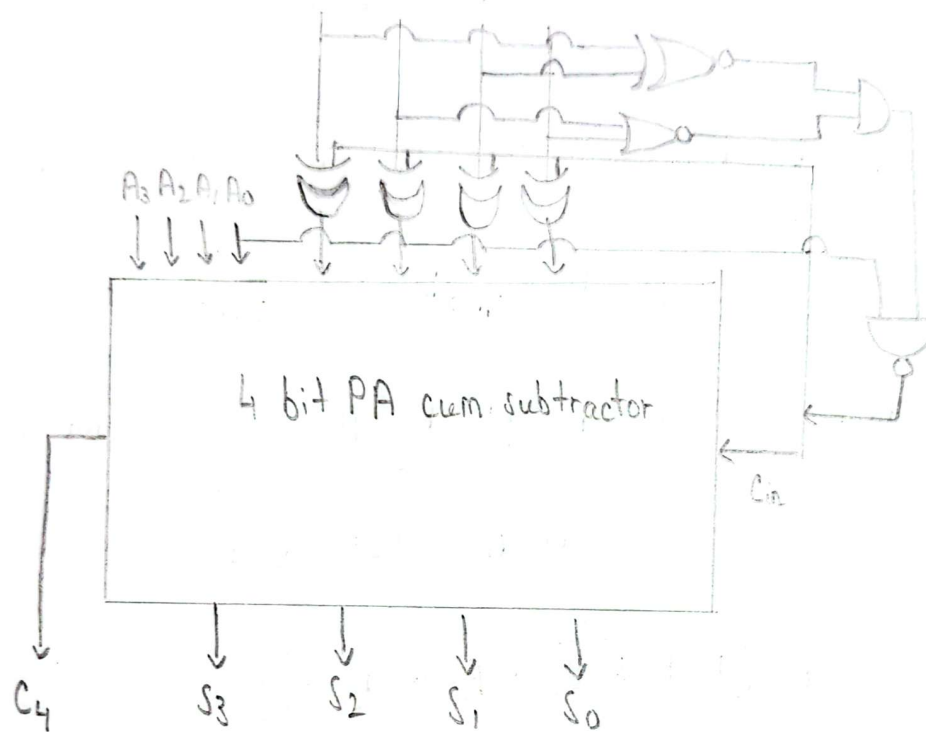
Numbers which is divisible by 10 are 0 and 10. [0000, 1010]

" " " odd has LSB as 1.

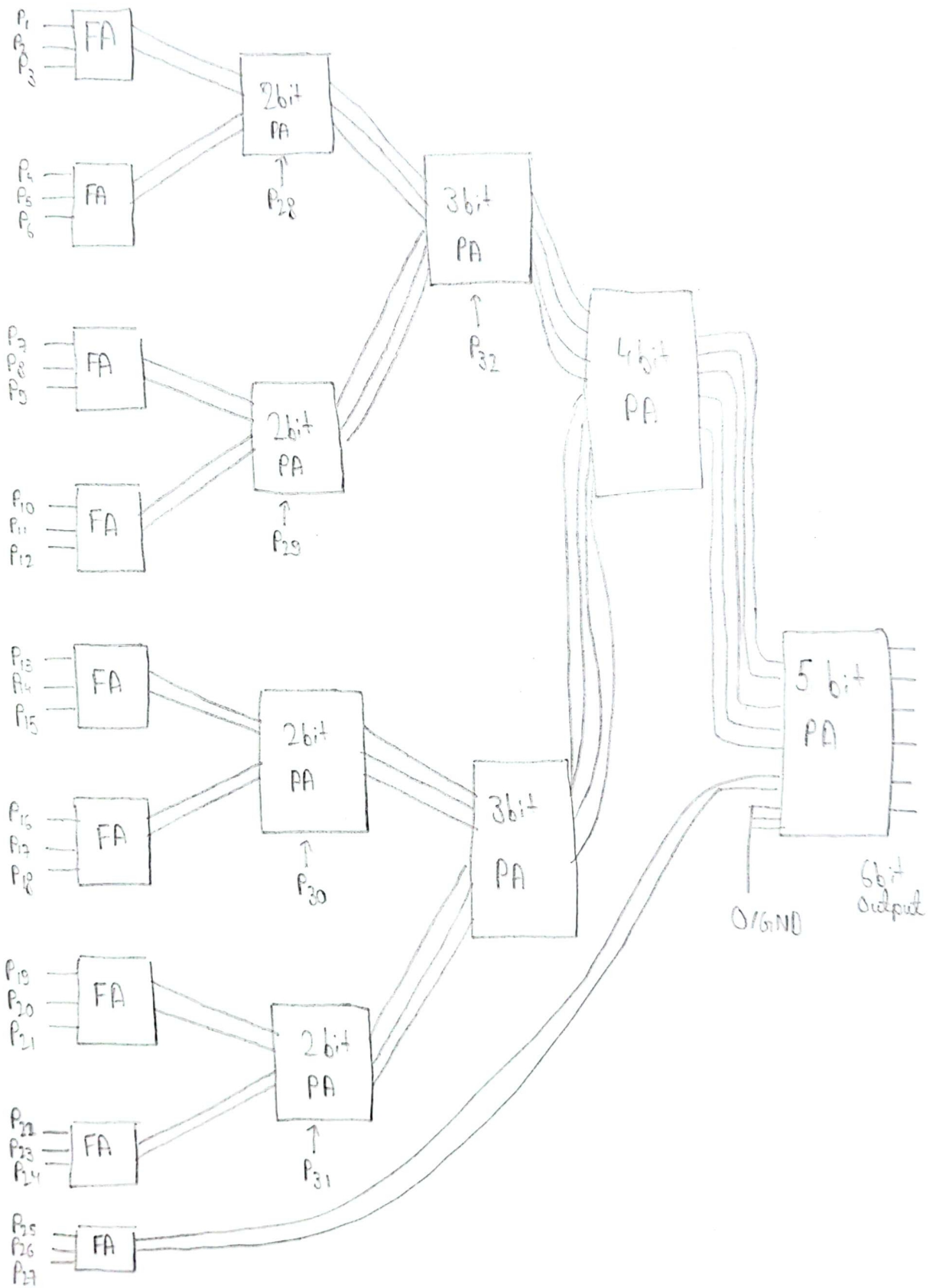
That is,  
~~∴ Select~~  $A_0 (\bar{B}_3 \bar{B}_2 \bar{B}_1 \bar{B}_0 + B_3 \bar{B}_2 B_1 \bar{B}_0)$

Here,

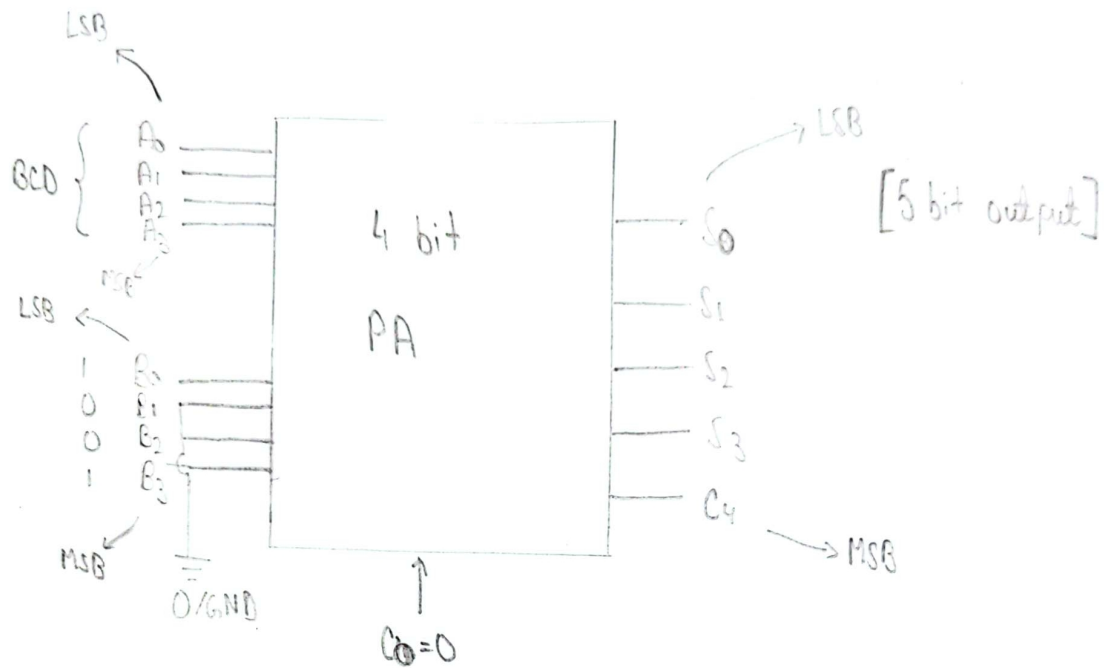
$B_3$  &  $B_1$  has a pattern &  $B_0$  and  $B_2$  has a pattern.



Ans. to the ques. no. 3



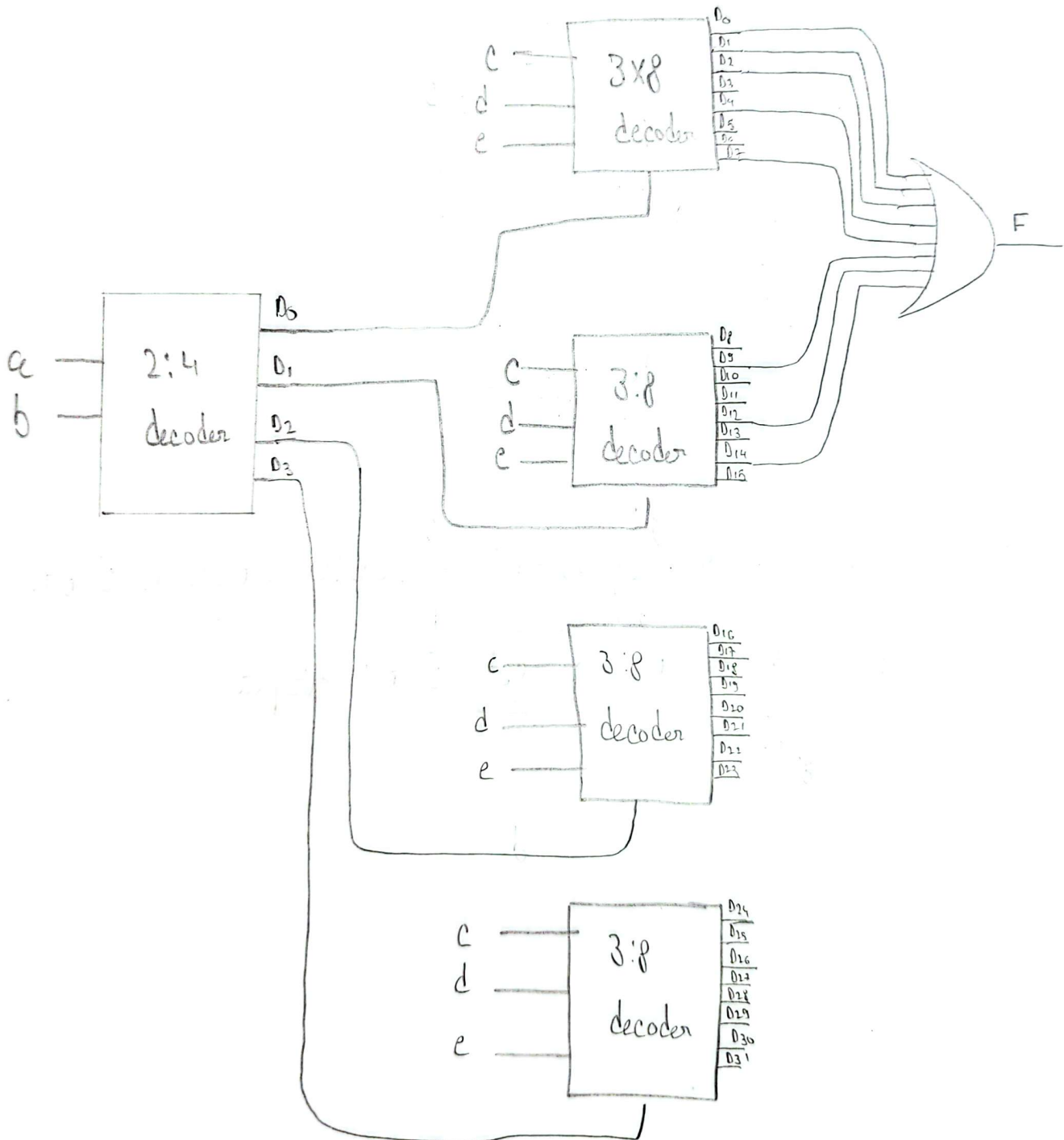
Ans. to the ques. no. 4



Excess 9 means  $(9+9) = 18 = 10010$  which is 5 bits to represent it. Thus we get 5 bit output.

Ans. to the ques. no. 5

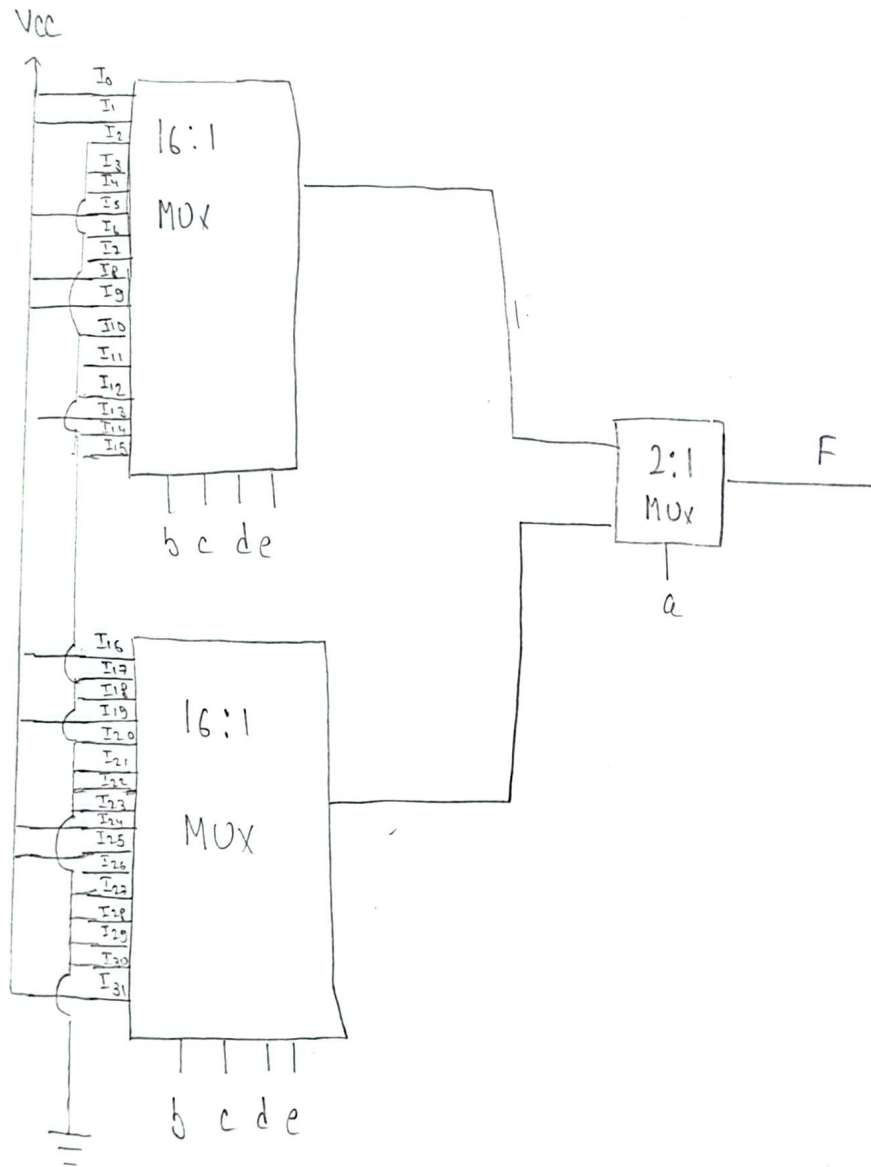
$$F(a, b, c, d, e) = \sum (0, 1, 2, 4, 7, 9, 12, 14)$$



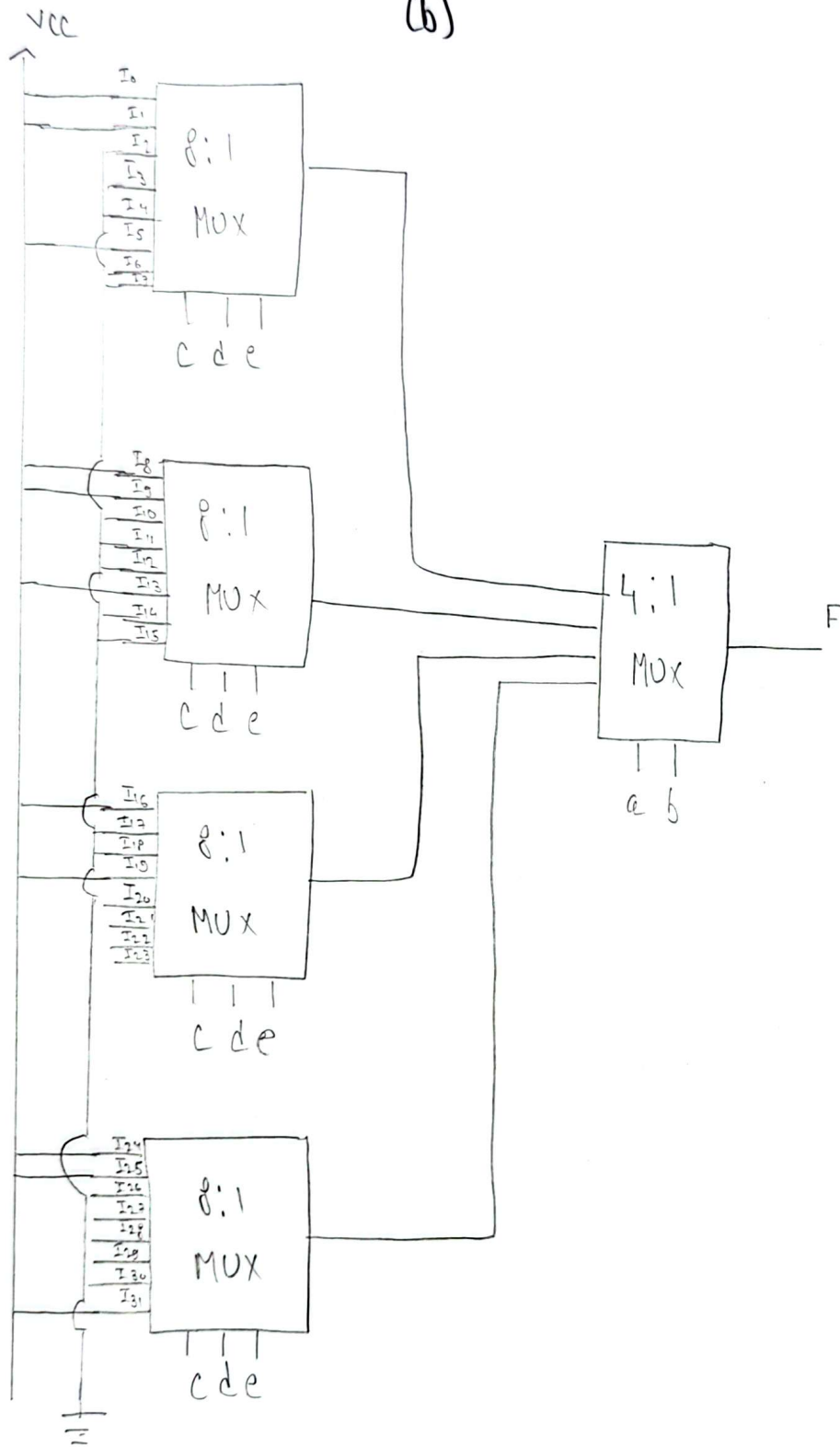
Ans. to the ques. no. 6

$$F(a, b, c, d, e) = \sum (0, 1, 5, 8, 9, 13, 16, 19, 24, 25, 31)$$

(a)



(b)

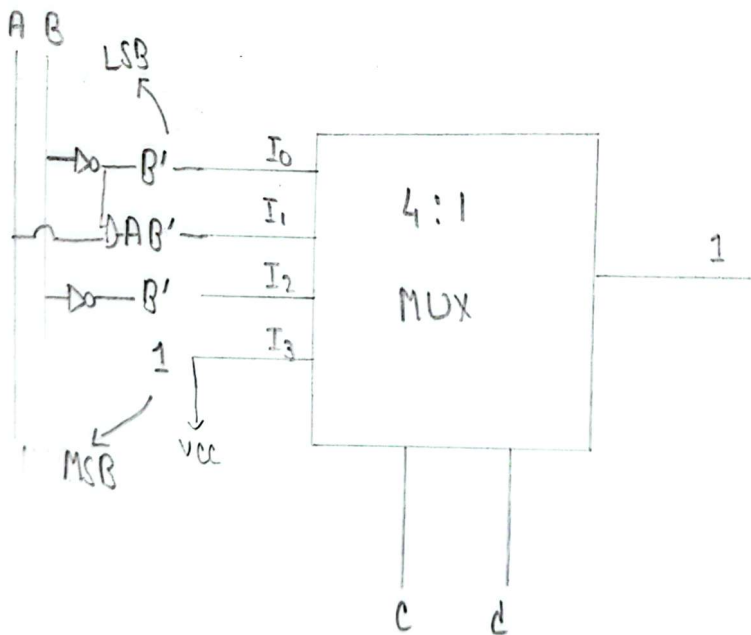


Ans. to the ques. no. 7

(b)

$$F(a,b,c,d) = \sum (0, 2, 3, 7, 8, 9, 10, 11, 15)$$

|        | $I_0$ | $I_1$ | $I_2$ | $I_3$ |
|--------|-------|-------|-------|-------|
| $A'B'$ | ①     | 1     | ②     | ③     |
| $A'B$  | 4     | 5     | 6     | ⑦     |
| $AB'$  | ⑧     | ⑨     | ⑩     | ⑪     |
| $AB$   | 12    | 13    | 14    | ⑮     |
|        | $B'$  | $AB'$ | $B'$  | 1     |





Ans. to the ques. no. 8

| Dec | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> | A <sub>1</sub> | A <sub>0</sub> | v |
|-----|----------------|----------------|----------------|----------------|----------------|----------------|---|
| 0   | 0              | 0              | 0              | 0              | X              | X              | 0 |
| 1   | 0              | 0              | 0              | 1              | 0              | 0              | 1 |
| 2   | 0              | 0              | 1              | 0              | 0              | 1              | 1 |
| 3   | 0              | 0              | 1              | 1              | 0              | 0              | 1 |
| 4   | 0              | 1              | 0              | 0              | 1              | 0              | 1 |
| 5   | 0              | 1              | 0              | 1              | 0              | 0              | 1 |
| 6   | 0              | 1              | 1              | 0              | 0              | 1              | 1 |
| 7   | 0              | 1              | 1              | 1              | 1              | 0              | 1 |
| 8   | 1              | 0              | 0              | 0              | 1              | 1              | 1 |
| 9   | 1              | 0              | 0              | 1              | 0              | 0              | 1 |
| 10  | 1              | 0              | 1              | 0              | 0              | 1              | 1 |
| 11  | 1              | 0              | 1              | 1              | 1              | 1              | 1 |
| 12  | 1              | 1              | 0              | 0              | 1              | 0              | 1 |
| 13  | 1              | 1              | 0              | 1              | 1              | 1              | 1 |
| 14  | 1              | 1              | 1              | 0              | 1              | 1              | 1 |
| 15  | 1              | 1              | 1              | 1              | 0              | 0              | 1 |

$A_1$

| $D_3 D_2$   | $D_1 D_0$ | $D_1' D_0'$ | $D_1 D_0$ | $D_1 D_0'$ |                    |
|-------------|-----------|-------------|-----------|------------|--------------------|
| $D_3' D_2'$ | X 0       | 1           | 3         | 2          |                    |
| $D_3' D_2$  | 4         | 5           | 7         | 6          | $D_3' D_2 D_1 D_0$ |
| $D_3 D_2$   | 10        | 13          | 15        | 14         |                    |
| $D_3 D_2'$  | 8         | 9           | 11        | 10         | $D_3 D_2 D_0'$     |

$D_3 D_2 D_1'$  (points to cell 1)  
 $D_1' D_0'$  (points to cell 13)  
 $D_3 D_2' D_1 D_0$  (points to cell 11)

$$A_1 = D_1' D_0' + D_3' D_2 D_1 D_0 + D_3 D_2' D_1 D_0 + D_3 D_2 D_0' + D_3 D_2 D_1'$$

$A_0$

| $D_3 D_2$   | $D_1 D_0$ | $D_1' D_0'$ | $D_1 D_0$ | $D_1 D_0'$ |             |
|-------------|-----------|-------------|-----------|------------|-------------|
| $D_3' D_2'$ | X 0       | 1           | 3         | 2          | $D_2' D_0'$ |
| $D_3' D_2$  | 4         | 5           | 7         | 6          |             |
| $D_3 D_2$   | 12        | 13          | 15        | 14         |             |
| $D_3 D_2'$  | 8         | 9           | 11        | 10         | $D_1 D_0'$  |

$D_3 D_2 D_1' D_0$  (points to cell 13)  
 $D_3 D_2' D_1$  (points to cell 11)

$$A_0 = D_1 D_0' + D_2' D_0' + D_3 D_2' D_1 + D_3 D_2 D_1' D_0$$

